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Rank-One Separable Potentials from Nucleon-Nucleon Phase Shifts (*).

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Summary. — We present a set of rank-one separable potentials for the nucleon-nucleon system corresponding to all partial waves up to $L=2$. The form factors of these interactions were obtained by using the solution of the inverse problem and by fitting their parameters to represent the Arndt-MacGregor-Wright nucleon-nucleon phase shifts. We have also studied the resulting potentials in computing two- and three-nucleon parameters.

1. – Introduction.

There exist at present several potential models which represent reasonably well the available nucleon-nucleon low-energy experimental data. Among them, the nonlocal separable potentials have shown to be very useful in different nuclear calculations. These separable interactions allow one to obtain the solution of both the direct and inverse scattering problem in a simple and elegant way. At the same time, those potentials provide a simple extension of the T-matrix off the energy shell and simplify considerably the treatment of the three-nucleon problem.

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The justification for using separable potentials has been extensively discussed in the literature [1-5].

Although in the case of a nonlocal potential it is not clear how to represent short-distance repulsions, it is possible to simulate those effects by using rank-two separable interactions. There is also an increasing interest in obtaining rank-one separable potentials with both attraction and repulsion included, which allows to describe the change of sign that appears in the experimental phase shifts as a function of the energy for some partial waves.

One clear advantage of using rank-one separable potentials is that GOURDIN and MARTIN [6], BOLSTERLI and MACKENZIE [7] and TABAKIN [8] have solved the inversion problem for them by using different methods and have given analytical expressions which relate the factors defining the separable interaction to the phase shifts and bound-state energies.

The main purpose of this paper is to present a set of rank-one separable potentials for the nucleon-nucleon system corresponding to all partial waves up to $L=2$. The analytical expression of these interactions were obtained by using the above-mentioned solution of the inversion problem and by fitting their parameters so as to reproduce the Arndt-MacGregor-Wright [9] nucleon-nucleon phase shifts. We have also studied the resulting potentials in computing two- and three-nucleon bound-state parameters.

In Sect. **2** we present a brief review of the inversion problem for separable potentials. Section **3** deals with the fitting procedure we have used and contains our results. Finally, in Sect. **4** some tests of the proposed potentials in connection with the deuteron and triton are discussed.

2. – The inversion problem.

The inversion problem for rank-one separable interactions has a closed solution in the sense that an analytical expression can be found relating the potential with the experimental two-body data, even when spinning particles

(1) P. NOYES: *Introductory Review in Three Body Problem*, edited by J. MCKEE and P. ROLPH (Amsterdam, 1970), p. 2.
(2) J. M. BLATT and V. F. WEISSKOPF: *Theoretical Nuclear Physics* (New York, 1952), p. 139.
(3) Y. YAMAGUCHI: *Phys. Rev.*, **95**, 1628 (1954).
(4) F. TABAKIN: *Phys. Rev.*, **174**, 1208 (1968).
(5) T. MONGAN: *Phys. Rev.*, **175**, 1260 (1968).
(6) M. GOURDIN and A. MARTIN: *Nuovo Cimento*, **8**, 699 (1958).
(7) M. BOLSTERLI and J. MACKENZIE: *Physics*, **2**, 141 (1965); M. BOLSTERLI: *Phys. Rev.*, **182**, 1095 (1969).
(8) F. TABAKIN: *Phys. Rev.*, **177**, 1443 (1969).
(9) R. A. ARNDT, M. H. MACGREGOR and R. M. WRIGHT: *Phys. Rev.*, **182**, 1714 (1969).

are considered. This is in contrast with the local case, where the potential can be expressed, at most, in terms of the solution of the Gel'fand-Levitan integral equation [10].

The separable type of interactions have been treated by several authors [6-8] using different approaches which lead essentially to the same results. We present here a summary of those results in order to establish the basic equations and conventions used in our fits.

We shall first consider coupled partial waves and show how to reduce the problem to the simpler uncoupled case.

The starting point is the partial-wave two-body Lipmann-Schwinger equation [*] for the nucleon-nucleon system:

$$T^{JSI}_{ll'}(k, k'; q^2) = V^{JSI}_{ll'}(k, k') + \frac{M}{\hbar^2} \sum_{l''=|J-S|}^{|J+S|} \int_0^\infty \frac{x^2 V^{JSI}_{ll''}(k, x)\, T^{JSI}_{l''l'}(k, k'; q^2)}{q^2 - x^2 + i0}\, \mathrm{d}x\,, \tag{1}$$

where J is the total angular momentum, I is the isospin of the system and l, l', l'' are orbital angular momenta. The partial-wave expansion of the potential appearing in eq. (1) is

$$\langle k', s', \nu' | V | k, s, \nu \rangle = \sum_{l,l',J} \langle k, l', s', \nu' | \eta | k, l, s, \nu \rangle V^{JSI}_{ll'}(k, k')\, \delta_{ss'} \delta_{\nu\nu'}\,, \tag{2}$$

where η is the generalized spherical harmonic [11] with analogus expressions for the T and S matrices.

In order to reduce the present problem to the case of uncoupled waves we can take profit of the freedom in the election of the potential structure. Thus we shall assume [7] that the same orthogonal matrix U

$$U(k) = \begin{pmatrix} \cos \varepsilon^\alpha(k) & \sin \varepsilon^\alpha(k) \\ -\sin \varepsilon^\alpha(k) & \cos \varepsilon^\alpha(k) \end{pmatrix} \tag{3}$$

diagonalizes simultaneously the S-matrix and the potential. In eq. (3) α stands for J, S, I.

Then we propose for the diagonal potential the following single-term separable expression:

$$\sum_{ll'} U^\alpha_{Ll}(k)\, V^\alpha_{ll'}(k, k')\, U^\alpha_{L'l'}(k') = \delta_{LL'} \frac{2\hbar^2}{M\pi} \sigma^\alpha_L g^\alpha_L(k) g^\alpha_L(k')\,. \tag{4}$$

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[10] R. NEWTON: *Scattering Theory of Waves and Particles* (New York, 1966), p. 613.

[*] We have used units such that $\hbar^2/M = 41.47\ \mathrm{fm}^2$.

[11] M. GOLDBERGER and K. WATSON: *Collision Theory* (New York, 1967), p. 347.

Here M is the nucleon mass, $2k = |k_1 - k_2|$ is the relative momentum and $\sigma_L^\alpha = \pm 1$ depending on L and α. As is usual, we call $g_L^\alpha(k)$ the form factor of the potential.

From eq. (1) we obtain

$$T_L^\alpha(k, k'; q^2) = \frac{2\hbar^2}{M\pi} \cdot \frac{\sigma_L^\alpha g_L^\alpha(k) g_L^\alpha(k')}{D_L^{\alpha+}(q^2)}, \tag{5}$$

where we have defined

$$D_L^{\alpha+}(q^2) = 1 + \sigma_L^\alpha \frac{2}{\pi} \cdot \int_0^\infty \frac{x^2[g_L^\alpha(x)]^2 \, \mathrm{d}x}{x^2 - q^2 - i0}. \tag{6}$$

Now the relation

$$\operatorname{Im} D_L^{\alpha+}(k) = \sigma_L^\alpha \cdot k \cdot [g_L^\alpha(k)]^2 \tag{7}$$

follows immediately.

From the Schrödinger equation and eq. (6) we see that a bound state for $E_B = -(\hbar^2/M)k_B^2$ exists whenever the equation

$$D_L^{\alpha+}(ik_B) = 0 \tag{8}$$

holds.

From the relation between the T-matrix and the phase shifts

$$T_L^\alpha(k) = -\frac{2\hbar^2}{M\pi} \exp[i\delta_L^\alpha(k)] \cdot \frac{\sin \delta_L^\alpha(k)}{k} \tag{9}$$

we obtain

$$\exp[2i\delta_L^\alpha(k)] = \frac{[D_L^{\alpha+}(k)]^*}{D_L^{\alpha+}(k)}, \tag{10}$$

which is a very useful formula that resembles the well-known relation between S-matrix and Jost functions. From eqs. (6) and (9) it follows

$$-\operatorname{tg} \delta_L^\alpha(k) = -\frac{k\sigma_L^\alpha[g_L^\alpha(k)]^2}{1 + \dfrac{2}{\pi}\sigma_L^\alpha \displaystyle\int_0^\infty \frac{x^2[g_L^\alpha(x)]^2}{x^2 - k^2}\,\mathrm{d}x}. \tag{11}$$

This tells us that the determination of the separable interaction has been reduced to the solution of an integral equation for the square of the form factor. In ref. [6] eq. (11) has been solved giving

$$[g_L^\alpha(k)]^2 = -\frac{\sin \delta_L^\alpha(k)}{k} \sigma_L^\alpha \exp\left[-\frac{1}{\pi}\int_0^\infty \frac{\delta_L^\alpha(k')}{k' - k}\,\mathrm{d}k'\right]\left[1 + \sum_{n=1}^{N} \frac{c_n}{k^{2n}}\right],$$

or

$$D_L^{\alpha+}(k) = \exp\left[-i\delta_L^\alpha(k)\right]\exp\left[-\frac{1}{\pi}\int_0^\infty \frac{\delta_L^\alpha(k')}{k'-k}\,\mathrm{d}k'\right]\left[1+\sum_{n=1}^{N}\frac{c_n}{k^{2n}}\right], \tag{12}$$

provided that the phase shifts satisfy the relation

$$\delta_L^\alpha(0)-\delta_L^\alpha(\infty) = N\pi\,, \tag{13}$$

where N stands for an integer.

From eq. (13) it is clear that $\sin\delta_L^\alpha(k)$ goes through zero at least $N-1$ times. Because $[g_L^\alpha(k)]^2$ has a constant sign, the zeros of $\sin\delta_L^\alpha(k)$ must coincide with those of the bracket in eq. (12). In ref. [6] it has been stated that a separable potential of the form (4) exists if and only if

a) $\sin\delta_L^\alpha(k)$ has N zeros and so all the constants c_n are well determined;

b) $\sin\delta_L^\alpha(k)$ has $N-1$ zeros and there is a bound state. In this case the condition (9) together with the position of the $N-1$ zeros determine the constants c_n.

If we call N_L the number of values k_{ci} where $\sin\delta_L^\alpha(k)$ changes sign and N_B the number of bound states (0 or 1), eq. (12) can be written as

$$[g_L^\alpha(k)]^2 = -\frac{\sigma_L^\alpha\sin\delta_L^\alpha(k)}{k}\exp\left[-\frac{1}{\pi}\int_0^\infty\frac{\delta_L^\alpha(k')}{k'-k}\,\mathrm{d}k'\right]\prod_{i=1}^{N_L}\frac{k^2-k_{ci}^2}{k^2}\cdot\left(\frac{k^2+k_B^2}{k^2}\right)^{N_B}, \tag{14}$$

where explicit use of expression (7) has been made.

Now eq. (13) is the statement of the modified Levinson theorem:

$$\delta_L^\alpha(0)-\delta_L^\alpha(\infty) = (N_B+N_L)\pi\,, \tag{15}$$

implying a restriction on the high-energy behaviour of the phase shifts. It is worth mentioning that an alternative way of solving the inversion problem is the one proposed by BOLSTERLI and MACKENZIE [7] and TABAKIN [8] which is based on the analytic properties of the function $D_L^{\alpha+}(k)$.

We conclude this Section remarking that the approach we have just described allows one to take into account possible changes of sign of the phase shifts using rank-one separable potentials and, in this way, to simulate the effects of short-range repulsions.

3. – Fits and results.

In order to take profit of eqs. (7), (12) and (13), obtained in the previous Section, relating the potential form factors with the experimental phase shifts, it is convenient to propose some suitable parametrization for the S-matrix.

We have choosen the Bargmann's ansatz [12]

$$S_L^\alpha(k) = \frac{P_L^\alpha(k)}{[P_L^\alpha(k)]^*} = \exp\left[2i\delta_L^\alpha(k)\right], \tag{16}$$

where $P_L^\alpha(k)$ is a polynomial in k. For a given partial wave, the analytical properties of the S-matrix and the low-energy behaviour of the phase shifts impose on the polynomials the following conditions:

$$P_L^\alpha(-k) = P_L^{\alpha *}(k) \tag{17}$$

and

$$\lim_{k\to 0} k^{2L+1} \operatorname{ctg} \delta_L^\alpha(k) = C \neq 0 \,. \tag{18}$$

From eq. (16) we can write

$$\delta_L^\alpha(k) = \frac{1}{2i} \sum_{j=1}^{N} \operatorname{Ln} \frac{k + i\alpha_j}{k - i\alpha_j}, \tag{19}$$

where α_j are the roots of $P_L^\alpha(k)$ and Ln indicates the principal branch of the logarithmic function.

Choosing $\delta_L^\alpha(\infty) = 0$, eq. (19) implies

$$\delta_L^\alpha(0) = \tfrac{1}{2}\pi(N^+ - N^-), \tag{20}$$

N^+ (N^-) being the number of α_j's with positive (negative) real part. The modified Levinson theorem (eq. (15)) and eq. (20) give an additional restriction on the polynomial $P_L^\alpha(k)$.

Using the above-mentioned machinery we have fitted the Arndt-MacGregor-Wright nucleon-nucleon phase shifts for laboratory kinetic energy from 0 to 460 MeV, for partial waves with $L = 2$.

A least-square procedure was used through a minimization program. For each partial wave we have used all the 35 data points available of the constrained solution of ref. [9].

In the case of S-waves, the scattering length, effective range and deuteron

[12] V. Bargmann: *Rev. Mod. Phys.*, **21**, 488 (1949).

TABLE I. – *Low-energy S-wave nucleon-nucleon parameters.* The value of the $^3S_1$ effective range is an output of our fit.

	Scattering length	Effective range	Deuteron binding energy
$^1S_0$	-23.679 fm	2.51 fm	—
$^3S_1$	5.379 fm	1.746 fm	-2.2245 MeV (c.m.)

binding energy were also supplied as input. The values taken for these parameters are summarized in Table I. It is worth mentioning that the order of the polynomial used depends on the particular partial wave considered; in all cases this order was less or equal than six. In Fig. 1 to 10 we display the difference between the fitted values and the central data points for the dif-

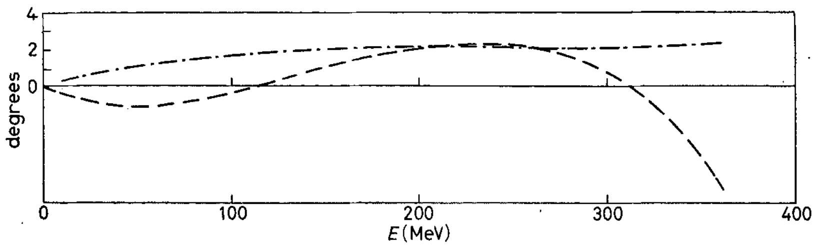

Fig. 1. – Fit to the $^1S_0$ phase shift. The curve marked —·—·— represents the experimental minimum uncertainties in degrees, given in ref. (9) and the points marked — — — are the differences between the fitted values and the central data points, also in degrees, both as a function of the laboratory kinetic energy.

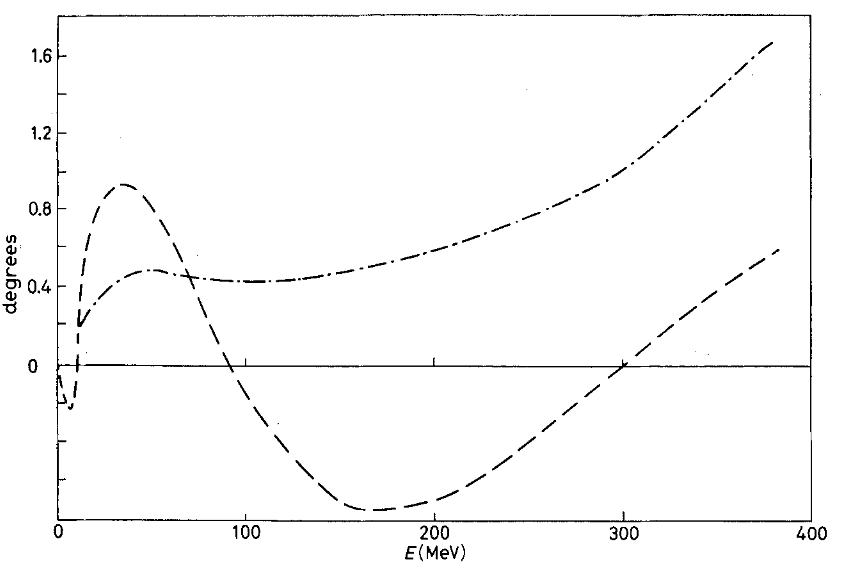

Fig. 2. – Fit to the $^3S_1$ phase shift. Description as given in Fig. 1.

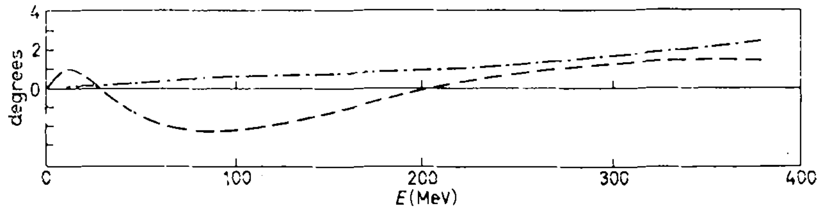

Fig. 3. – Fit to the $^1P_1$ phase shift. Description as given in Fig. 1.

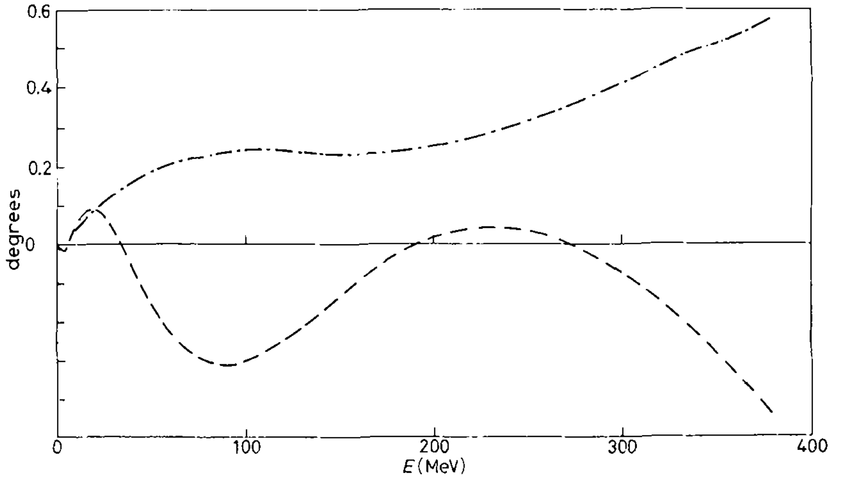

Fig. 4. – Fit to the $^3P_0$ phase shift. Description as given in Fig. 1.

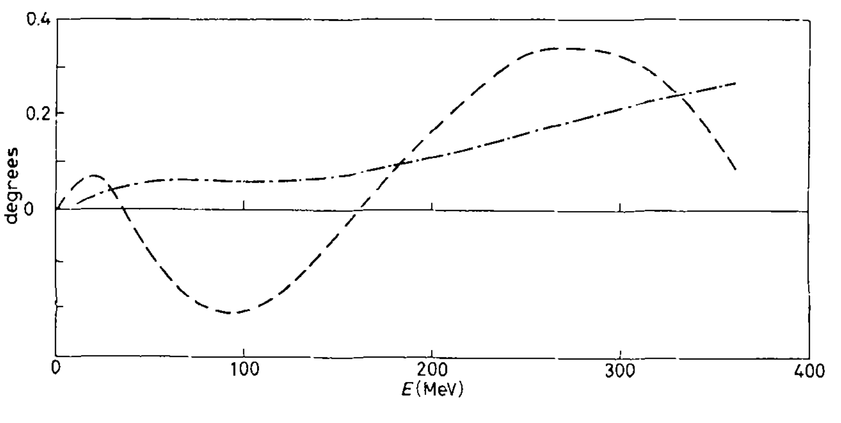

Fig. 5. – Fit to the $^3P_1$ phase shift. Description as given in Fig. 1.

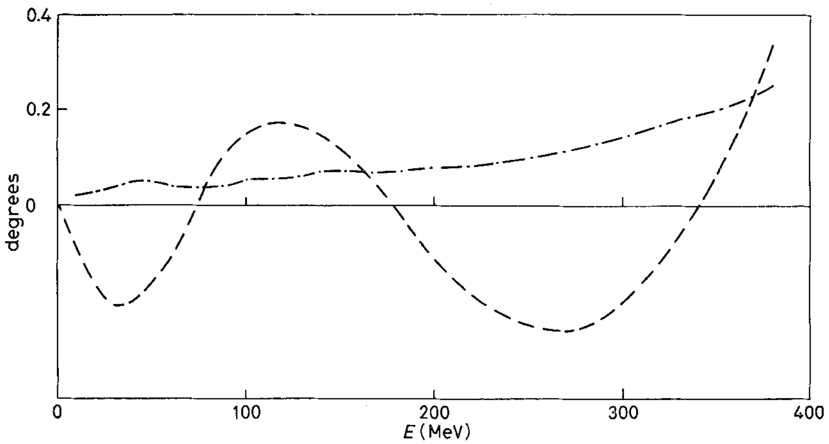

Fig. 6. – Fit to the $^3P_2$ phase shift. Description as given in Fig. 1.

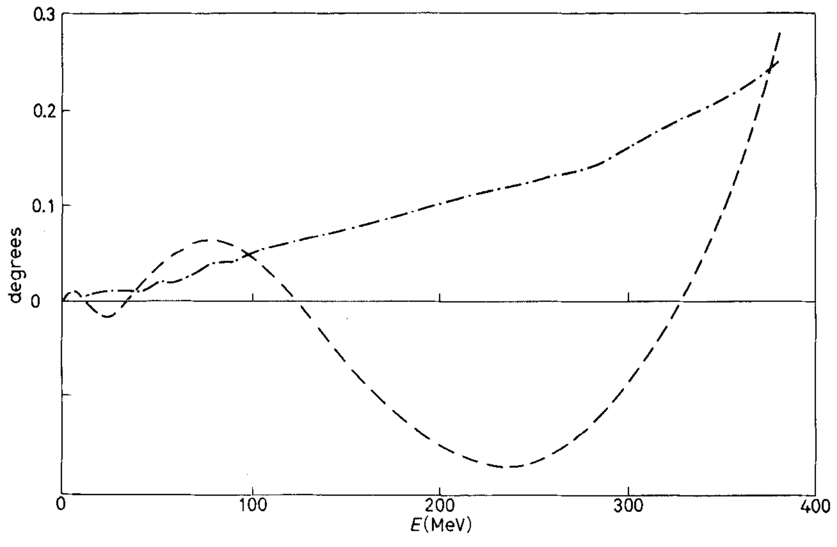

Fig. 7. – Fit to the $^1D_2$ phase shift. Description as given in Fig. 1.

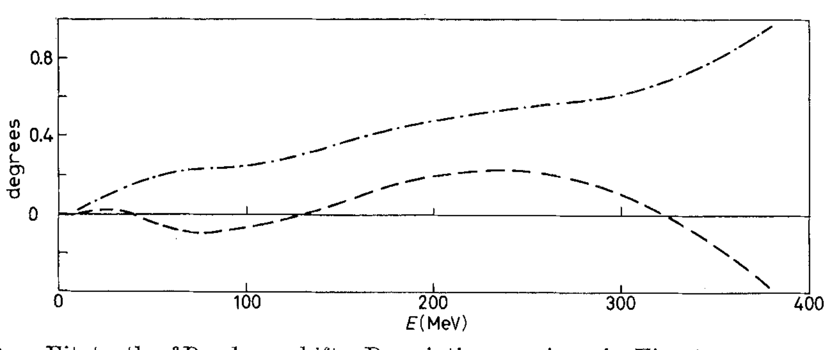

Fig. 8. – Fit to the $^3D_1$ phase shift. Description as given in Fig. 1.

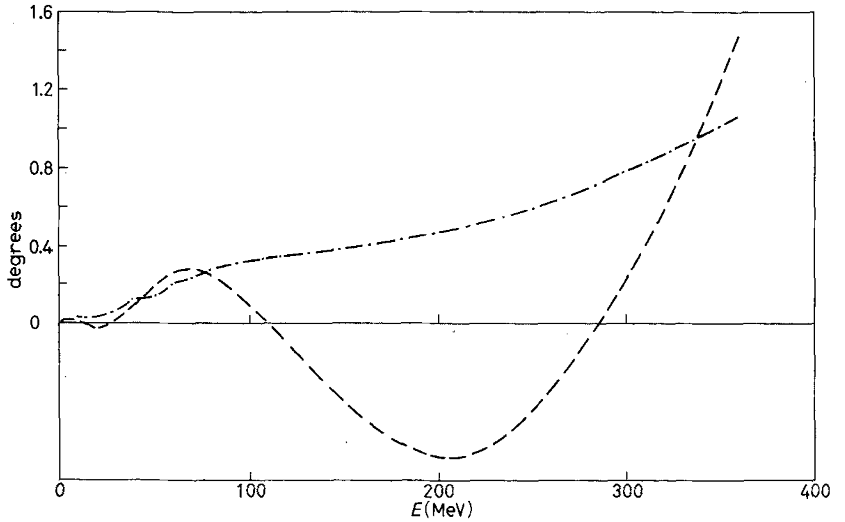

Fig. 9. – Fit to the $^3D_2$ phase shift. Description as given in Fig. 1.

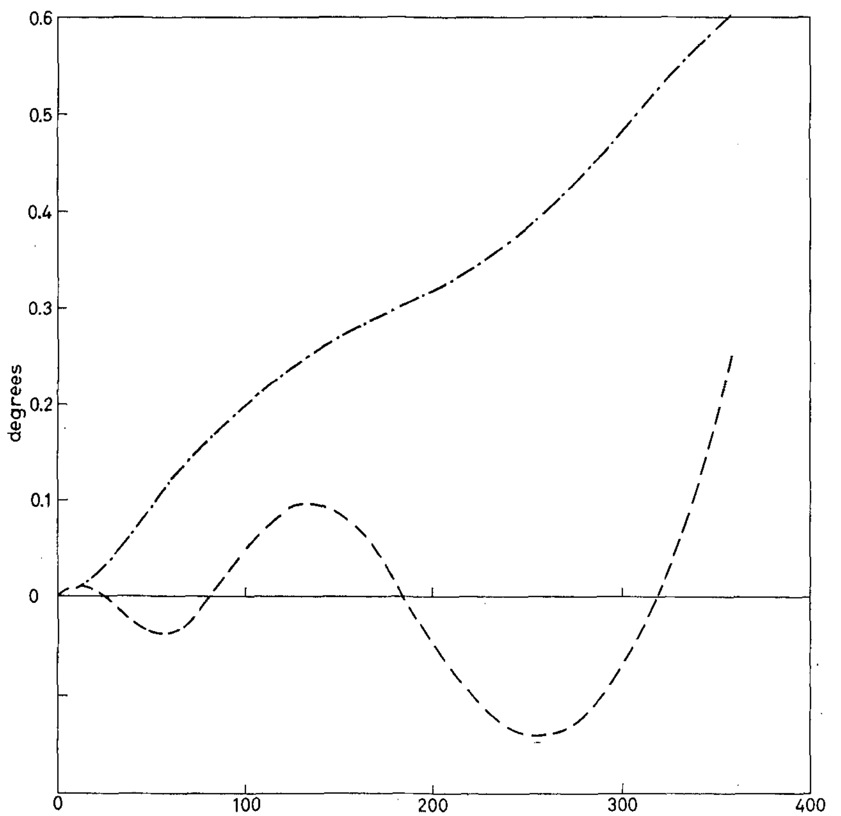

Fig. 10. – Fit to the $^3D_3$ phase shift. Description as given in Fig. 1.

ferent partial waves together with the uncertainties of the corresponding experimental phase shifts given in ref. [9] as a function of the laboratory kinetic energy. The fact that these errors are lower-bound estimates implies that the χ^2 values of the fit are meaningless and for that reason we do not quote them.

The mixing parameter $\varepsilon^\alpha(k)$ for the ${}^3S_1 - {}^3D_1$ coupled waves needs a different handling. Due to the fact that it is an even function of k, and goes to zero with k as k^2, it was fitted by using for $\operatorname{tg} \varepsilon^\alpha(k)$ a polynomial of the form

$$\operatorname{tg} \varepsilon^\alpha(k) = k^2(a + bk^2 + ck^4 + dk^6)\,, \tag{21}$$

where the resulting parameters are given in Table II.

TABLE II. *Parameters defining* $\operatorname{tg} \varepsilon^\alpha$ *given by eq.* (22).

a	b	c	d
0.0610	-0.0208	0.00795	-0.00073

With the ansatz (16) it is easy to find an analytical expression for the form factor. Replacing (19) in (12) we find for the function

$$D_L^{\alpha+}(k) = \left(\frac{k^2 + k_B^2}{k^2}\right)^{N_B} \prod_{i=1}^{N_L} \frac{k^2 - k_{ci}^2}{k^2} \prod_{j=1}^{N^+} \frac{k}{k + i\beta_j} \prod_{j=1}^{N^-} \frac{k + i\gamma_j}{k}\,, \tag{22}$$

where β_j, $-\gamma_j$ are the polynomial roots with positive or negative real part respectively.

From this last equation one inmediatly obtains the general expression for

TABLE III. – *Fitted parameters involved in the definition of the form factors given in eq.* (24) *for the ten partial waves considered.*

	σ_L^α	λ	k_c^2 (fm^{-2})	N_L	γ (fm^{-2})	η	μ (fm^{-2})	ε (fm^{-4})	ϱ (fm^{-2})	N_B
${}^1S_0$	-1	1.182	3.738	1	0.8165	1	-11.851	36.815	—	0
${}^3S_1$	-1	2.133	3.534	1	1.662	1	-10.783	99.593	—	0
${}^1P_1$	$+1$	10	—	0	0.5529	0	1	7.658	—	0
${}^3P_0$	-1	100	2.478	1	0.1745	1	-0.2860	27.325	9.995	1
${}^3P_1$	$+1$	14.284	—	0	0.1257	1	-13.995	—	—	0
${}^3P_2$	-1	7.276	—	0	1.534	0	1	4.907	30.854	1
${}^1D_2$	-1	1.618	—	0	0.02331	0	1	1.234	34.198	1
${}^3D_1$	$+1$	1.386	—	0	0.08251	1	1.623	2.442	—	0
${}^3D_2$	-1	1.259	—	0	0.06322	1	0.4369	1.228	—	0
${}^3D_3$	-1	0.02949	—	0	0.0494	1	-1.499	3.647	—	0

the form factors

$$g_L^\alpha(k) = \frac{\lambda(k^2 - k_c^2)^{N_L} k^L}{[(k^2+\gamma)(\eta k^4 + \mu k^2 + \varepsilon)(k^2+\varrho)^{N_E}]^{\frac{1}{2}}} . \tag{23}$$

Table III gives the quantities λ, k_c, N_L, γ, η, μ, ε, ϱ and N_E for all the partial waves up to $L = 2$, obtained using the mentioned procedure.

4. – Tests and conclusions.

In the fits described in the previous Section neither the deuteron D-state probability nor its quadrupole moment were used as input. Thus, the computation of their values can be considered as a first test of the potentials.

For the D-state probability we have obtained $P_D = 4.76\,\%$, in accordance with the current estimates of this value. We consider this a satisfactory feature, because it has been remarked [13,14] the importance of the correctness of the P_D value in connection with the treatment of the three-nucleon problem, as well as in nuclear-matter calculations.

Through the use of numerical methods we have computed the deuteron quadrupole moment Q from the well-known equation

$$Q = \frac{1}{10}\sqrt{2}\left\{\int_0^\infty \mathrm{d}k\, \omega_2^D(k)\left[k\frac{\mathrm{d}\omega_0^D}{\mathrm{d}k} - k^2\frac{\mathrm{d}^2\omega_0^D}{\mathrm{d}k}\right]\right\} - \\ - \frac{1}{20}\left\{\int_0^\infty \mathrm{d}k\left[6(\omega_2^D(k))^2 + \left(k\frac{\mathrm{d}\omega_0^D}{\mathrm{d}k}\right)^2\right]\right\} . \tag{24}$$

The obtained result is $Q = 0.027\ \text{fm}^2$. We have also computed the value of the S-wave r.m.s. radius $\sqrt{\langle r^2\rangle} = 3.86\ \text{fm}$.

The evaluated quadrupole moment is certainly too small compared with the experimental value. This can be attributed in part to the low-energy behaviour of the experimental mixing parameter, where errors are significant.

As a test of the off-the-energy-shell behaviour of our potential model we have plotted in Fig. 11, as an example, the reaction matrix corresponding to the ${}^1S_0$ partial wave, given by the expression

$$R_{J,J}^\alpha(k, k'; k'^2) = \frac{2\hbar^2}{M\pi}\cdot\frac{g_J^\alpha(k)\, g_J^\alpha(k')}{|D_J^{\alpha+}(k')|\cos\delta_J^\alpha(k')} ,$$

([13]) A. C. PHILLIPS: *Nucl. Phys.*, **107** A, 209 (1968).

([14]) T. BRADY, M. FUDA, E. HARMS, J. S. LEVINGER and R. STAGAT: *Phys. Rev.*, **186**, 1069 (1969).

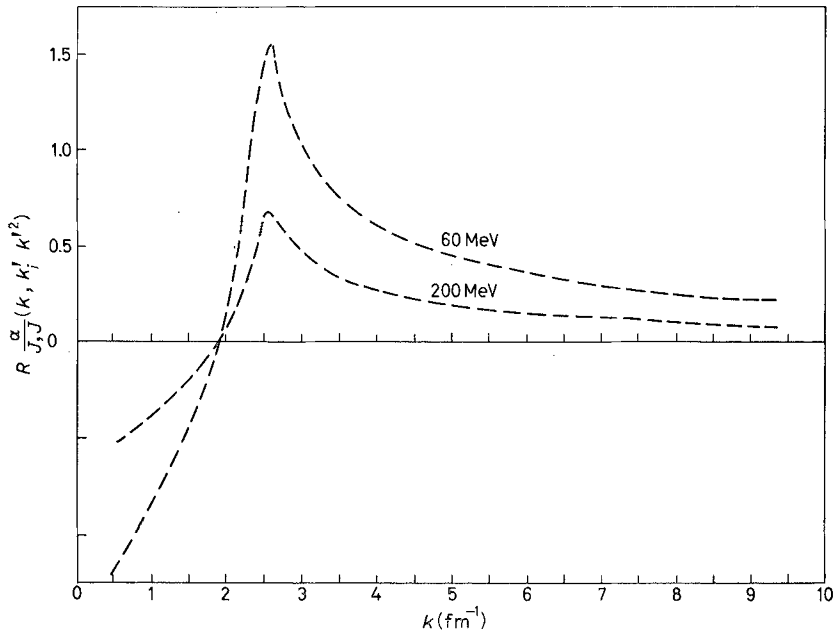

Fig. 11. – The reaction matrix $R^{\alpha}_{J,J}(k, k'; k'^2)$ for the ${}^1S_0$ partial wave with k as a parameter.

for different values of E. The result is in complete agreement with the R-matrix for other potentials [15].

For our case the « repulsive » bump is found to be around 2.6 fm^{-1} which can be interpreted as a repulsive core of 0.4 fm.

A study of the deuteron wave functions in configuration space, computed following the method of HURWITZ and ZWEIFEL [16], shows the characteristic features of a short-range repulsion.

The off-energy-shell behaviour of a two-body interaction can be tested in a more severe fashion through the calculation of three-nucleon parameters. Using our ${}^1S_0$ and ${}^3S_1$ separable potentials in the Faddeev-Lovelace equations we have computed the triton binding energy, with the result $E_B = -8.25$ MeV, in good agreement with the experimental value.

At this point we comment the fact that even if the form factors of this potential involve square roots and short-range repulsion in a single term, we have not found spurious bound states, *i.e.* no ghost appears [17].

The fits we have presented result in a set of nucleon-nucleon separable

(15) D. SPRUNG and M. SRIVASTAVA: *Nucl. Phys.*, **149** A, 113 (1970).
(16) H. HURWITZ and P. ZWEIFEL: *MTAC*, **10**, 140 (1956).
(17) M. BANDER: *Phys. Rev.*, **138**, B 322 (1965).

potentials whose analytical expressions involve only a few parameters, and reproduce fairly well the experimental phase shifts, and the off-shell behaviour of the two-nucleon amplitude as indicated by the excellent results obtained in the three-nucleon system. Even though we are aware of the fact that the form factors we have used may give rise to a bound state in the continuum (at the energy where the phase shift changes sign) we regard the whole picture as satisfactory and conclude that our potentials can be used to describe the nucleon-nucleon interaction in nucleon structure calculations.

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● RIASSUNTO (*)

Si presenta un insieme di potenziali separabili di primo rango per il sistema nucleone-nucleone corrispondente a tutte le onde parziali sino a $L=2$. Si sono ottenuti i fattori di forma di queste interazioni usando la soluzione del problema inverso ed approssimando i loro parametri per rappresentare gli spostamenti di fase nucleone-nucleone di Arndt-MacGregor-Wright. Si sono studiati anche i potenziali risultanti nel computo dei parametri di due e tre nucleoni.

(*) *Traduzione a cura della Redazione.*

Разделяемые потенциалы из нуклон-нуклонных фазовых сдвигов.

Резюме (*). — Мы предлагаем систему разделяемых потенциалов для нуклон-нуклонной системы, соответствующей всем парциальным волнам вплоть до $L=2$. Форм-факторы этих взаимодействий получаются с помощью использования решения обратной проблемы и подгонки рассматриваемых параметров, чтобы воспроизвести нуклон-нуклонные фазовые сдвиги. Мы также исследовали результирующие потенциалы при вычислении двух- и трех-нуклонных параметров.

(*) *Переведено редакцией.*